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Access to Parkinson's disease treatments in the Middle East: a multi-national survey of availability, affordability, and utilization patterns

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ABSTRACT

Background: Parkinson's disease (PD) is an increasing public health challenge in the Middle East, yet regional data on access to therapies and services remain limited. Understanding the gaps in PD treatment availability, affordability, and prescription frequency is necessary for improving care delivery and regional policy development.

Objectives: Evaluating the availability, affordability, and utilization of disease-related pharmacological treatments, surgical, device-assisted therapies, and rehabilitative services across multiple Middle Eastern countries.

Methods: A cross-sectional, web-based survey was distributed to physicians with expertise in movement disorders across participating countries in the Middle East. The survey assessed three domains of availability, cost, and utilization of key medications, surgical interventions, and specialized PD services.

Results: Thirty-one respondents from nine Middle Eastern countries, including 44% movement-disorder specialists. Medication availability was 65.4%, affordability 68.6%, and prescription 52% across the included countries. Levodopa was nearly universally available (98%), whereas newer dopamine agonists, MAO-B inhibitors, and COMT inhibitors were available inconsistently. Advanced therapies demonstrated disparities; botulinum toxin and deep-brain stimulation were available in high-income countries but not frequently used, while infusion therapies were mostly unavailable. Telemedicine uptake was moderate; genetic testing was cost-restricted. Although general rehabilitation was available, PD-specific services were limited. Affordability constrained equitable care. Qatar had the highest reported prescription frequency of PD medications among the countries.

Conclusions: The survey highlights disparities in access to essential PD therapies across the Middle East. Variable availability, costs, and limited specialized services prevent optimal care, urging the need for stronger regional policies, improved insurance coverage, and expanded clinical training to improve equity in PD management.

1. Introduction

Parkinson's disease (PD) is the second most common progressive neurodegenerative disorder, affecting over 10 million people worldwide and manifesting with diverse motor and non-motor symptoms that significantly impair quality of life [1]. While dopaminergic replacement therapy remains the main treatment strategy, advanced therapeutics are often required for complex or treatment-resistant cases [2].

Over the last three decades, PD rates have increased in Middle Eastern countries due to an aging population, a sedentary lifestyle, and

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industrialization [3]. Despite therapeutic advances, significant disparities in access to PD treatments remain between high- and low- to middle-income countries. According to World Bank income classifications, Saudi Arabia, Oman, the United Arab Emirates, Qatar, Bahrain, and Kuwait, which are often regarded as Gulf Cooperation Council (GCC) countries, are high-income; Iran and Iraq are upper-middle income; and Jordan, Syria, Lebanon, and Yemen are lower-middle or low income [4].

Notable disparities in socioeconomic conditions, healthcare infrastructure, and pharmaceutical policies lead to significant differences in the availability and affordability of PD treatment across Middle Eastern countries [5]. A global survey demonstrated that although levodopa was widely available, other medications and advanced therapies were less accessible in upper-middle and lower-income countries [6]. These socioeconomic inequalities prevent ideal PD care by limiting treatment access and medication adherence [7].

In the Middle East, there is inconsistent healthcare financing, a limited number of movement disorders specialists, and an uneven distribution of advanced therapy centers [8]. Therefore, many patients in these countries struggle with high treatment costs associated with their income and limited insurance coverage, resulting in harder access to care and lower use of recommended treatments [8].

Despite growing recognition of PD as a global health priority, access to essential therapies remains variable across regions, particularly in low- and middle-income settings. Recent policy frameworks from the World Health Organization, emphasize that barriers to treatment access are multifactorial, including limitations in availability, affordability, health system infrastructure, workforce capacity, and policy implementation. These barriers are often conceptualized within systems-based models that highlight interconnections between supply chains, financing mechanisms, and service delivery [9].

Beyond patient and provider-level perspectives, access to PD therapies can also be evaluated through complementary approaches such as pharmacy-based availability surveys, assessments of alignment between national essential medicines lists, standard treatment guidelines, and health insurance coverage policies. However, such comprehensive evaluations remain limited in the Middle East, where data on the real-world accessibility, affordability, and utilization of PD treatments are scarce.

According to these challenges, a comprehensive understanding of the availability, affordability, and utilization patterns of PD treatments in the Middle East is crucial. Such evidence can inform health policy, guide resource allocation, and identify existing disparities within care pathways and lead to better and equal access to healthcare globally. Thus, the present study aims to assess access to pharmacological, device-based, and rehabilitative PD therapies across multiple Middle Eastern countries.

2. Methods

We conducted this cross-sectional study from March to September 2025 to evaluate the availability, affordability, and utilization of PD treatments across Middle Eastern countries. We developed a structured web-based survey instrument comprising items addressing access to pharmacological therapies, advanced device-aided treatments, and specialized PD care services and rehabilitative tools. The International Parkinson's and Movement Disorder Society (MDS) provided the survey distribution platform and enabled distribution through the Middle East Working Group network of MDS. The survey link was disseminated electronically to neurologists and movement disorder specialists involved in PD care across the region.

Participants from 12 Middle Eastern countries (Egypt, Bahrain, Iran, Iraq, Jordan, Kuwait, Lebanon, Qatar, Saudi Arabia, Syria, Turkey, and the United Arab Emirates) were invited through professional networks, national neurology societies, and the regional section of the International Parkinson MDS. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki [10] and adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies.

Board-certified neurologists and movement disorder specialists with more than five years of experience in managing patients with movement disorders, actively practicing in Middle Eastern countries, were invited to participate in the study and asked to complete a structured questionnaire. The questionnaire which is accessible in the supplementary file, was designed by senior movement disorder specialists and consisted of all the pharmacological treatment and advanced therapies in the management of PD patients followed by inquiring three key items of Availability (Access through pharmacies and hospital presence), Affordability (Cost of treatment relative to income and insurance coverage), Prescription rate (Frequency of prescription among the patients).

We used a standardized ordinal scale for each of the items in the survey. Availability of each treatment and specialized care was rated on a five-point scale as follows: 1 = not available, 2 = occasionally available, 3 = sometimes available, 4 = mostly available, and 5 = always available. Affordability was assessed using a three-point scale: 1 = not affordable, 2 = partially affordable, and 3 = completely affordable. Prescription frequency was evaluated using a five-point scale, with 1 = not prescribed, 2 = rarely prescribed, 3 = sometimes prescribed, 4 = frequently prescribed, and 5 = continuously prescribed. This scoring system was utilized by all of the participants to allow equal comparison across the countries.

The aforementioned therapies included:

- Levodopa-based formulations (Levodopa/carbidopa (formulation of 100/25, 250/25 or other), levodopa/benserazide (formulation of 100/25, 200/50), extended-release forms, subcutaneous foslevodopa, foscarbidopa)
- Dopamine agonists (Pramipexole, ropinirole, rotigotine patch, bromocriptine, cabergoline, lisuride, pergolide, piribedil)
- Monoamine oxidase B (MAO-B) inhibitors (Selegiline, rasagiline, safinamide)
- Catechol-*O*-methyltransferase (COMT) inhibitors (Entacapone, tolcapone, opicapone)
- N-Methyl-D-aspartic acid (NMDA) antagonists (Amantadine, extended release amantadine)
- Anticholinergics (Trihexyphenidyl, procyclidine, benztropine)
- Cholinesterase inhibitors (Donepezil, rivastigmine)
- Atypical antipsychotics (Quetiapine, clozapine)

And a group of other drugs associated with symptom relief in PD patients, such as benzodiazepine and non-benzodiazepine sedatives, and laxatives.

Advanced therapies such as Surgical treatments (Deep Brain Stimulation surgery, Ablative surgery), Continuous apomorphine both subcutaneous and pump, Levodopa carbidopa intestinal gel (LCIG), asynchronous telemedicine (meaning not necessarily online at same time; WhatsApp or similar apps), synchronous telemedicine (meaning online and connecting at the same time; virtual visits), Rehabilitation (Physical therapy, occupational therapy, speech therapy) in general and PD-specific methods, Complementary therapies (Including alternative medical options) and special tools designed specifically for PD patients.

The questionnaire was distributed to participating physicians via a Google Form link. After receiving all the forms, responses were extracted from the Google Form application and merged together to use for analysis purposes.

Data were analyzed using R (R Foundation for Statistical Computing, version 4.2.0) with the tidyverse. Variables related to availability, affordability, and prescription frequency were systematically identified and extracted alongside country identifiers. For each country, mean scores for these measures were calculated across respondents for each therapy, excluding missing values. The data were reshaped into long

format to facilitate cross-country comparisons. Country-level sample sizes were computed and merged with the aggregated dataset to aid interpretation. Drug labels were standardized for clarity. Results were visualized as heatmaps using ggplot2 with the viridis color scale to represent mean scores [11].

3. Results

A total of 31 respondents from 9 of the 12 Middle Eastern countries completed the survey. The mean age of the participants was 49.74 ± 8.62 years. The mean duration of experience in visiting PD patients and prescribing medication was 16.35 ± 8.06 years. Additionally, 74.2% of the respondents were male. Most of the respondents were from Iran (n = 21), followed by Turkey and the United Arab Emirates (UAE) with 2 respondents each, Iraq, Egypt, Saudi Arabia, Kuwait, Jordan, and Qatar.

Approximately 44% identified as movement-disorder specialists, while the remainder were general neurologists or other health-care workers involved in the care of patients with PD.

Clinical experience varied widely, with respondents reporting a minimum of 5 to 30 years of practice in neurological care.

Table 1 outlines prevalence data, national population figures, and the distribution of neurologists, movement-disorder clinics, and specialists across countries. The presented data were obtained from publicly accessible databases and online resources. Country-level demographic data were retrieved from World Population Review. Information on neurological workforce metrics was obtained from the Atlas of MS and the International Parkinson's and Movement Disorder Society Specialist Finder tool. Data on PD prevalence were extracted from Our World in Data, based on estimates from the Global Burden of Disease study.

A) Medical therapies

Across all medications, the mean availability score was 65.4%, indicating suboptimal consistency, particularly for advanced and adjunctive therapies. Countries such as Jordan and Qatar showed higher availability scores across most drug classes, whereas others demonstrated gaps, particularly for advanced formulations and non-dopaminergic agents. The mean affordability score was 68.6%, with newer or specialty treatments disproportionately less affordable. Almost all medications available in Qatar were noted as completely affordable, followed by the UAE being partially affordable, while other countries demonstrated heterogeneous affordability among different medication classes. Prescription patterns demonstrated a mean score of 52%.

Table 1
Population size, Parkinson's disease prevalence, and neurologist workforce across the Middle-Eastern countries included in the survey, the sources for all numerical values are provided in the table footnote.

Countries	2025 Population[1]	Number of neurologists per 100,000 people[2]	Movement Disorder Specialists[3] (n.)	Prevalence of PD per 100,000[4]
Egypt	118,366,000	3	3	62.7
Iran	92,417,700	1.9	5	84.6
Iraq	47,020,800	N/M	1	45.8
Jordan	11,520,700	N/M	–	37.3
Kuwait	5,026,080	1.1	–	50
Qatar	3,115,890	10	1	32.1
Turkey	87,685,400	3.6	5	104.5
Saudi Arabia	34,566,300	1	–	46.7
United Arab Emirates	11,346,000	1.5	2	47.6

[1] http://worldpopulationreview.com/countries
[2] https://atlasofms.org/chart
[3] https://mds.movementdisorders.org/specialist-finder
[4] https://ourworldindata.org/grapher/parkinsons-disease-prevalence-ihme

Prescription patterns were generally consistent with medication availability, and Qatar exhibited the highest prescription frequency of PD therapies across all surveyed countries. This moderate level suggests cautious or constrained prescribing practices, likely influenced by limited availability and cost barriers (Fig. 1).

Levodopa-based formulations remained the main disease management across all surveyed countries, with an availability rate of approximately 98%. Access to dopamine receptor agonists was more variable; pramipexole and bromocriptine were the most consistently available agents, whereas other agents in this class were largely inaccessible outside the GCC countries, such as Kuwait, Qatar, and the UAE.

Among MAO-B inhibitors, rasagiline and selegiline were available in approximately 75% of surveyed centers, while safinamide was accessible in only 25%. COMT inhibitors demonstrated heterogeneous availability. Entacapone was comparatively more accessible (40%), whereas opicapone and tolcapone showed lower availability (25% each), with access reported in only 15% of centers. Amantadine and anticholinergic agents (trihexyphenidyl and biperiden) remained affordable and commonly utilized adjunctive therapies, particularly in lower-income countries.

B) Advanced therapies

Respondents reported substantial heterogeneity in the availability, affordability, and prescription of advanced Parkinson's disease therapies, including device-aided treatments, surgical interventions, telemedicine modalities, botulinum toxin, and genetic testing in clinical practice. Botulinum toxin demonstrated consistently high availability across countries (predominantly always available), with partial affordability and a prescription frequency of 74%. Surgical interventions showed variable availability; deep brain stimulation (DBS) was slightly more accessible (66.4%) than ablative procedures (65.8%), particularly in the UAE and Qatar. However, both were associated with low affordability and prescription rates that did not parallel their reported availability.

Advanced infusion therapies exhibited the greatest access limitations. LCIG was largely unavailable or only occasionally available (36.6%) and infrequently prescribed (32.2%), while continuous subcutaneous apomorphine infusion (CSAI) showed higher but still limited availability (66.4%). Both therapies ranked lowest in affordability.

Telemedicine services demonstrated moderate availability (67%) and comparatively better affordability than pharmacological and surgical options, although utilization varied considerably across countries, with higher uptake in Saudi Arabia and the UAE. Genetic testing was reported as moderately to highly available in several countries (70%); nevertheless, affordability barriers and lower prescription frequency indicate limited integration into routine clinical practice.

Overall, treatments were generally less affordable than they were available, and often prescribed less frequently than their availability would suggest, suggesting financial constraints and practical challenges that limit access to advanced PD care across the region (Fig. 2).

C) Rehabilitative therapies

General rehabilitation services, including physiotherapy, speech therapy, swallow therapy, and occupational therapy, were widely available; however, PD-specific equivalents were consistently less available and less frequently prescribed, indicating a gap between general and disease-specific care. Specialized PD services, such as movement disorder clinics, showed moderate availability (60%) but limited affordability, while PD nurses and supportive general practitioners were scarce (38%) across most Middle Eastern countries. Psychological and cognitive services, including dietitian support and access to psychologists and psychiatrists, were generally available but underprescribed, with cognitive behavioral therapy (60.6%) and biofeedback (45%) demonstrating particularly low availability and affordability.

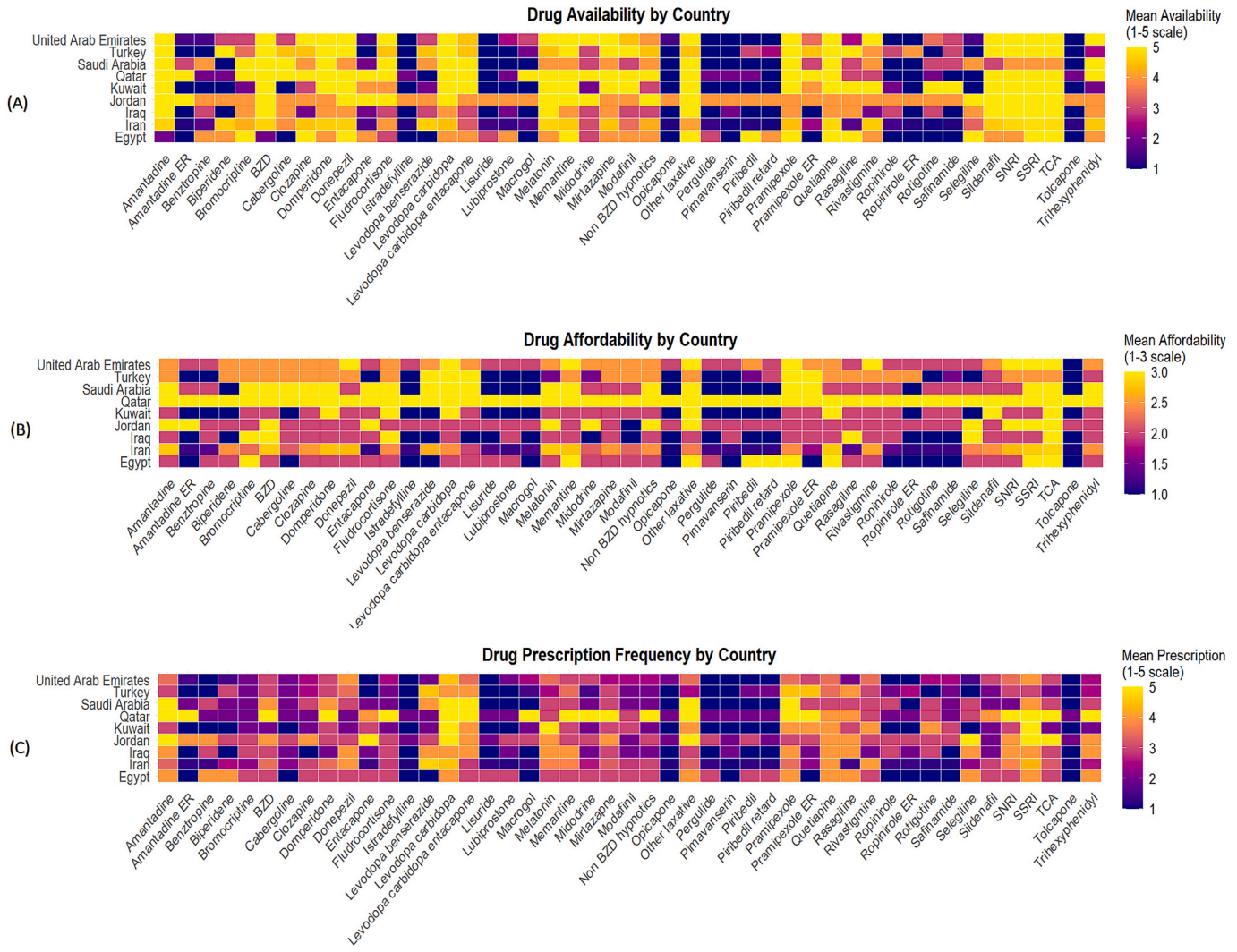


Fig. 1. Heatmap visualization of PD pharmacological treatments across nine Middle Eastern countries (United Arab Emirates, Turkey, Saudi Arabia, Qatar, Kuwait, Jordan, Iraq, Iran, and Egypt). Panel (A) illustrates the mean availability of PD medications on a 1–5 scale, panel (B) depicts affordability on a 1–3 scale, and panel (C) shows prescription frequency on a 1–5 scale. Color gradients represent increasing levels of availability, affordability, and prescribing frequency, respectively, highlighting inter-country variability across different classes of PD therapies. ER: Extended-release, BZD: Benzodiazepines, SNRI: Serotonin-Norepinephrine Reuptake Inhibitor, SSRI: Selective Serotonin Reuptake Inhibitor, TCA: Tricyclic antidepressants.

Innovative and adjunctive therapies, including sensory cueing, adaptive devices, sport therapy, and complementary interventions, were largely unavailable or only occasionally available and infrequently prescribed. Affordability scores were uniformly lower than availability scores across all service categories, representing a consistent barrier to utilization. Country-level comparison showed higher availability and prescription of PD-specific rehabilitation services in Qatar, with Jordan and the UAE also showing stronger access profiles, whereas Iran exhibited good availability of general rehabilitation and psychological services but limited PD-specific and innovative therapies. Turkey demonstrated strong availability of core rehabilitation services but limited specialization, while Egypt, Iraq, Kuwait, and Saudi Arabia were characterized by restricted access to PD-specific, multidisciplinary, and innovative rehabilitative care (Fig. 3). In general, prescriptions were often less frequent than the services were actually available, especially for psychological and specialized care, limiting access to comprehensive PD rehabilitation in these countries.

4. Discussion

This multinational survey provides one of the first regional overviews of treatment access for PD across the Middle East, revealing substantial disparities in the availability, affordability, and utilization of core PD therapies and services.

Overall, our findings indicate that therapeutic access across the Middle East is moderate, with widespread availability of standard dopaminergic medications but considerably restricted access to advanced or device-based interventions. The observed pattern highlights the ongoing structural and economic inequities within the region, particularly the divide between GCC countries and non-GCC nations, which is likely shaped by variations in healthcare investment, insurance, and reimbursement policies, and the availability of specialized movement-disorders infrastructure.

Consistent with reports from other low- and middle-resource settings, including recent data from Africa and Latin America, our findings highlight structural barriers that prevent optimal PD management and contribute to inequities in patient outcomes [12,13]. Although levodopa-based therapies were widely available, affordability and insurance coverage varied significantly. Inconsistent access may compromise symptom control and worsen disability progression. This trend aligns with Global Burden of Disease findings, which link national development levels to health outcomes, demonstrating that middle- and

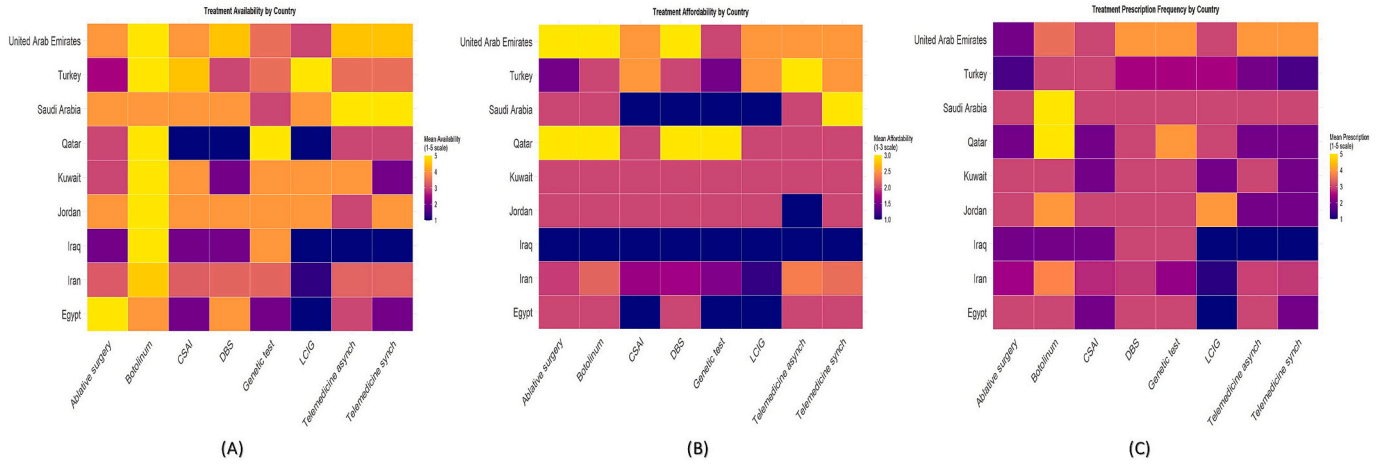


Fig. 2. Heatmap representation of advanced PD therapies across nine Middle Eastern countries. Panel (A) illustrates the mean availability of advanced treatments (ablative surgery, botulinum toxin, continuous subcutaneous apomorphine infusion, deep brain stimulation, genetic testing, Levodopa Carbidopa intestinal gel, asynchronous and synchronous telemedicine) on a 1–5 scale. Panel (B) depicts affordability on a 1–3 scale, and panel (C) shows prescription or utilization frequency on a 1–5 scale. Color gradients reflect increasing levels of availability, affordability, and clinical use, highlighting substantial inter-country variability in access to advanced PD care across the region. CSAI: continuous subcutaneous apomorphine infusion, DBS: deep brain stimulation, LCIG: Levodopa Carbidopa intestinal gel, asynch: asynchronous, synch: synchronous.

high-income countries generally achieve more equitable and comprehensive care than lower-income settings [14].

Evidence from Latin American countries further reflects the challenges identified in our Middle Eastern survey. PD patients face barriers to healthcare access across Central and South America. Participants described major gaps, such as delayed diagnosis due to limited provider knowledge, scarce multidisciplinary services, and inconsistent medication availability for those living in rural regions or unable to afford private care [15].

Findings from several availability studies conducted across African countries show that access to PD medications is highly limited, particularly within public pharmacies where treatments are generally more affordable. These assessments consistently report that dopamine agonists and anticholinergic agents, followed to a lesser extent by levodopa, tend to be the medications most commonly accessible for PD patients. The evidence highlights persistent supply limitations in low-resource settings [16,17]. A study of the WHO African Region found gaps in the inclusion of PD medications in national essential medicines lists, with 11% of countries lacking any key PD medicines and limited availability of recommended formulations, and identified poor prioritization of PD, shortages of trained healthcare workers, and financing barriers as major contributors to limited access to PD treatment in the region [18]. Qualitative data from Kenya indicated that PD medications were largely unavailable and unaffordable, leading to a substantial financial burden, poor symptom control, and reduced quality of life for patients [19]. The availability of rehabilitative therapies in Ghana is mostly limited to tertiary centers. While physiotherapy is available across most healthcare levels, speech therapy and occupational therapy remain limited [20]. Moreover, limitations in access to both motor and non-motor PD medications, with drugs for dementia, urinary symptoms, hallucinations, and autonomic dysfunction available in fewer than half of the surveyed nations, while antidepressants and gastrointestinal treatments were comparatively easier to obtain [13]. These restrictions mean that many patients rely on less expensive but less effective therapies. In addition to medication shortages, African healthcare systems also face severe workforce constraints, with very few neurologists and a scarcity of PD nurses and specialized clinics. Telemedicine remains underutilized among clinicians and patients in this region. Financial barriers also play a significant role, as most healthcare expenses are paid out-of-pocket, and insurance coverage is minimal [13].

A recent review from Central Asia and Transcaucasian countries similarly highlights major gaps in specialized PD care, including a shortage of neurologists with PD expertise and an absence of PD-trained nurses, speech, and occupational therapists. Although most PD medications are technically available, high costs, particularly for dopamine agonists and MAO inhibitors, significantly limit access. Advanced therapies such as DBS are offered only in Kazakhstan's capital, leaving much of the region underserved, given its vast geography and the need for multiple centers to ensure adequate follow-up [21].

Another review evaluating the cost of PD in several European countries shows that the condition imposes a significant financial burden on both patients and society. As the disease progresses and symptoms worsen, the associated healthcare and caregiving costs increase substantially [22].

Evidence from available advanced treatments across the Middle East demonstrates significant gaps in the availability of specialized movement-disorder care and advanced therapies such as DBS across the Middle East. Although PD is highly common in the region, specialized services remain limited [23]. Only a handful of tertiary centers currently offer DBS or other advanced treatments, and access is hindered by shortages of trained neurologists, inadequate public and provider awareness, and cultural factors that delay help-seeking. Countries such as Saudi Arabia, Kuwait, and the UAE are making progress in establishing comprehensive movement-disorders programs, most of the region still lacks the infrastructure, workforce, and patient-education systems required for equitable access to advanced PD care [23].

A recent review highlights international variation in PD prescribing patterns. While levodopa remains the most consistently used therapy worldwide, the use of ergot-derived dopamine agonists has declined due to safety concerns, with non-ergot agonists becoming more common. Substantial differences existed across countries in the use of COMT inhibitors, MAO-B inhibitors, amantadine, and anticholinergics, and prescribing choices were strongly influenced by both patient-related factors and prescriber preferences [24], underscoring persistent disparities in therapeutic access and practice patterns internationally, that support our observation of Middle Eastern countries exhibiting similar inconsistencies in availability, affordability, and utilization of PD medications.

This study has several limitations. Only 9 of 12 targeted Middle Eastern countries participated, limiting generalizability. Data were based on physician-reported perceptions rather than objective verification, introducing potential reporting bias and failing to capture within-

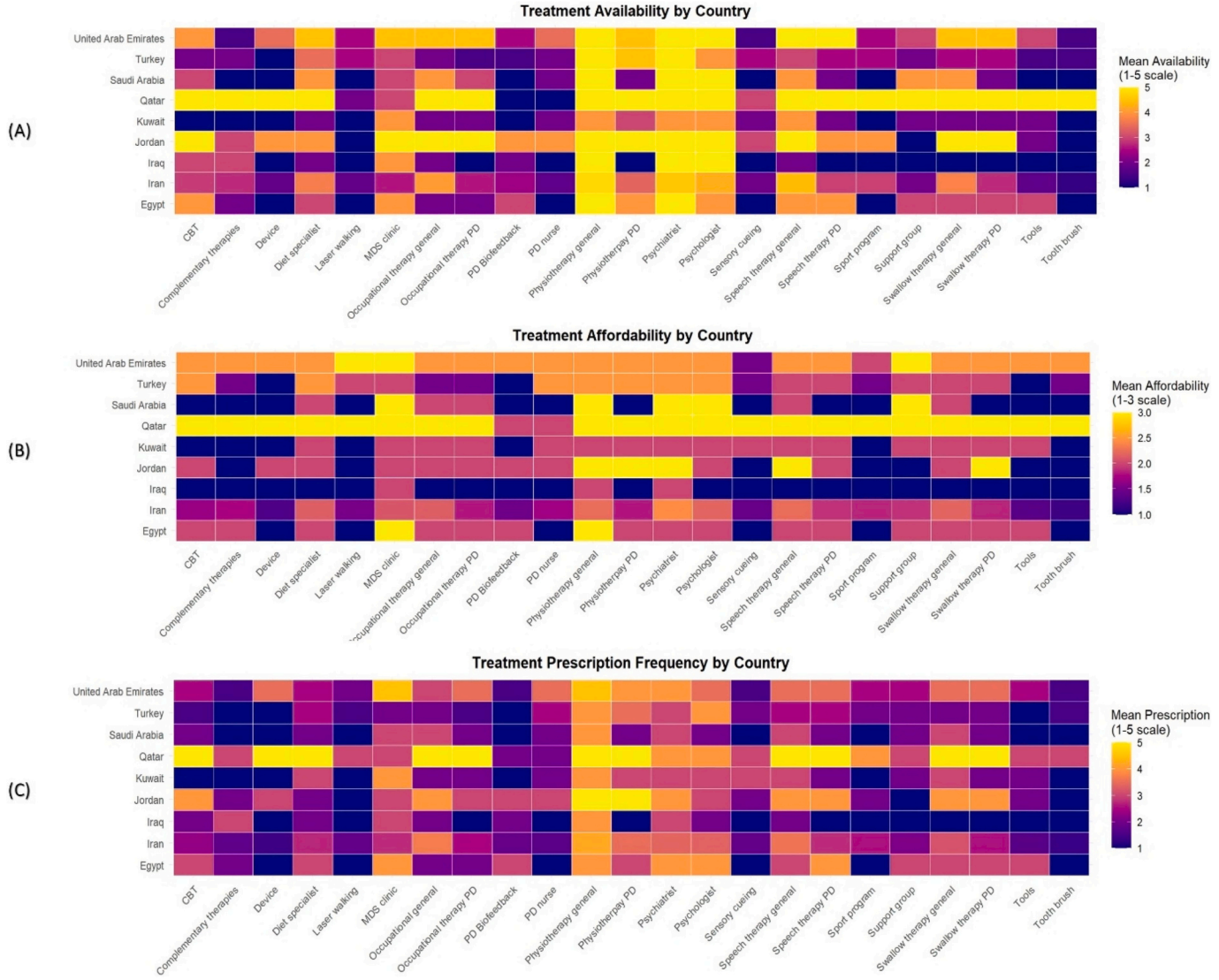


Fig. 3. Heatmap depiction of rehabilitative and supportive therapies for PD across nine Middle Eastern countries. Panel (A) shows the mean availability of rehabilitative services (including physiotherapy, occupational therapy, speech and swallow therapy, psychological and psychiatric care, PD nursing, cognitive–behavioral therapy, and supportive programs) on a 1–5 scale. Panel (B) illustrates affordability on a 1–3 scale, while panel (C) presents prescription or utilization frequency on a 1–5 scale. Color gradients indicate increasing levels of access and use, underscoring substantial inter-country disparities in the provision of comprehensive rehabilitative PD care across the region. CBT: Cognitive behavioral therapy, MDS: Movement disorder specialist, PD: Parkinson's disease.

country variability. Uneven respondent distribution, particularly the higher representation from Iran, may have influenced regional estimates. Affordability was assessed qualitatively without standardized economic metrics, and patient-reported outcomes or real-world adherence were not evaluated.

Overall, this multinational survey highlights disparities in access to PD treatments across the Middle East. While foundational dopaminergic medications are widely accessible, advanced and device-assisted therapies, as well as essential rehabilitative services, remain markedly limited across much of the region, particularly outside wealthier Gulf countries. Socioeconomic status, national health expenditure, and reimbursement structures emerged as key determinants of treatment availability and affordability, contributing to uneven care pathways between high-income and lower-income Middle Eastern states.

These findings emphasize the need for coordinated regional and national strategies to strengthen equitable access to PD care. Priority actions include expanding insurance coverage and subsidy mechanisms for high-cost therapies, establishing regional procurement and price-stabilization initiatives, supporting local pharmaceutical production, and investing in specialized training programs for neurologists and allied health professionals in movement disorders. Importantly, improving access in lower-resource settings, beyond the region's wealthy countries, must be central to future policy efforts. By implementing these measures, the Middle East can progress towards a more balanced, evidence-based, and inclusive PD care framework, ultimately enhancing treatment outcomes and quality of life for all individuals with PD in the region.

Ethical compliance statement

Ethics approval was provided by the Shahid Beheshti University of Medical Sciences ethics committee. No informed patient consent was necessary for this work. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this work is consistent with those guidelines.

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The authors have no financial conflicts of interest to disclose for the previous 12 months.

CRediT authorship contribution statement

Mehri Salari: Writing – review & editing, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Shivam Mittal:** Writing – review & editing, Resources. **Zakiyah Aldaajani:** Writing – review & editing, Resources. **Hanan Khalil:** Writing – review & editing, Resources. **Masoud Etemadifar:** Writing – review & editing, Methodology. **Murat Gültekin:** Writing – review & editing, Resources. **Firas Alfwaress:** Writing – review & editing, Resources. **Mudassar Iqbal Arain:** Writing – review & editing, Resources. **Sevim Soleimani:** Writing – original draft, Visualization, Software, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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During the preparation of this manuscript, the authors used Grammarly to edit and improve the language. Following the use of this tool/service, the authors carefully reviewed and revised the text as needed and take full responsibility for the content of the article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.prdoa.2026.100482.

Data availability

All relevant data have been included in the manuscript and supplementary material; further inquiries can be directed to the corresponding author and will be provided upon reasonable request.

References

[1] Y. Luo, L. Qiao, M. Li, X. Wen, W. Zhang, X. Li, Global, regional, national epidemiology and trends of Parkinson's disease from 1990 to 2021: findings from the Global Burden of Disease Study 2021, Front. Aging Neurosci. 16 (2024) 1498756.
[2] T. Foltynie, V. Bruno, S. Fox, A.A. Kühn, F. Lindop, A.J. Lees, Medical, surgical, and physical treatments for Parkinson's disease, Lancet 403 (10423) (2024) 305–324.
[3] S. Safiri, M. Noori, S.A. Nejadghaderi, S.E. Mousavi, M.J.M. Sullman, M. Araj-Khodaei, et al., The burden of Parkinson's disease in the Middle East and North Africa region, 1990-2019: results from the global burden of disease study 2019, BMC Public Health 23 (1) (2023) 107.
[4] Eric Metreau Key, Shwetha Grace Eapen, World Bank Blogs [Internet], Available from: https://blogs.worldbank.org/en/opendata/understanding-country-income–world-bank-group-income-classifica, 2025.
[5] A. Assa'ad, A.Y. Hershko, C. Irani, M. Mahdavinia, D.A. Khan, J.A. Bernstein, Health disparities in the Middle East: representative analysis of the region, J. Allergy Clin. Immunol. Glob. 4 (1) (2025) 100350.
[6] Z.H.K. Goh, J.L.Y. Cheong, C. Marras, C.M. Tanner, M. Kasten, A.D. Korczyn, et al., Surveying global availability of Parkinson's disease treatment, J. Parkinsons Dis. 12 (3) (2022) 1023–1034.
[7] G. Gil, M.H.S. Tosin, H.B. Ferraz, The impact of the socioeconomic factor on Parkinson's disease medication adherence: a scoping review, Arq. Neuropsiquiatr. 82 (2) (2024) 1–8.
[8] H. Khalil, L.M. Chahine, J. Siddiqui, M. Salari, S. El-Jaafary, Z. Aldaajani, et al., Parkinson's disease in the Middle East, North Africa, and South Asia: consensus from the International Parkinson and Movement Disorder Society Task Force for the Middle East, J. Parkinsons Dis. 10 (2) (2020) 729–741.
[9] Organization WH, Intersectoral Global Action Plan on Epilepsy and Other Neurological Disorders 2022–2031, Geneva: CC BY-NC-SA 3.0 IGO, 2023.
[10] World Medical Association, Declaration of Helsinki: ethical principles for medical research involving human subjects, JAMA 310 (20) (2013) 2191–2194.
[11] R Core Team, R: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing, Vienna, Austria, 2022. URL, https://www.R-project.org/.
[12] D.J. Kim, A.M. Rodriguez-Salgado, J.J. Llibre-Rodriguez, I. Acosta, A.L. Sosa, D. Acosta, et al., Burden of Parkinsonism and Parkinson's disease on health service use and outcomes in Latin America, J. Parkinsons Dis. 13 (7) (2023) 1199–1211.
[13] E. Hamid, B.A. Ayele, D.G. Massi, S. Ben Sassi, H. Tibar, E.E. Djonga, et al., Availability of therapies and Services for Parkinson's disease in Africa: a continent-wide survey, Mov. Disord. 36 (10) (2021) 2393–2407.
[14] Global, regional, and national burden of Parkinson's disease, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016, Lancet Neurol. 17 (11) (2018) 939–953.
[15] C. Jeyachandran, C. Spooner, Salgado AMR, M. Prina, J. Rhee, J.J. Llibre-Guerra, et al., Barriers and facilitators to accessing healthcare for people with Parkinson's disease in Latin America: a qualitative study, Health Expect. 28 (4) (2025) e70380.
[16] J. Mokaya, C.L. Dotchin, W.K. Gray, J. Hooker, R.W. Walker, The accessibility of Parkinson's disease medication in Kenya: results of a national survey, Mov. Disord. Clin. Pract. 3 (4) (2016) 376–381.
[17] N.U. Okubadejo, O.O. Ojo, K.W. Wahab, S.A. Abubakar, O.Y. Obiabo, F.K. Salawu, et al., A Nationwide survey of Parkinson's disease medicines availability and affordability in Nigeria, Mov. Disord. Clin. Pract. 6 (1) (2019) 27–33.
[18] N. Fothergill-Misbah, R. Cataldi, M. Stothard, N. Chowdhary, N. Okubadejo, B. Cappello, et al., Parkinson disease treatments on national essential medicines lists, African region, Bull. World Health Organ. 103 (10) (2025) 626–634.
[19] N. Fothergill-Misbah, J. Hooker, J. Kwasa, R. Walker, Access to medicines for Parkinson's disease in Kenya: a qualitative exploration, Mov. Disord. Clin. Pract. 11 (11) (2024) 1373–1378.
[20] M.W. Agoriwo, C. Joseph, E. Franzén, M. Unger, Structure and process of Parkinson's disease rehabilitation in the southern sector of Ghana: a cross-sectional survey, J. Eval. Clin. Pract. 31 (3) (2025) e14169.
[21] R. Kaiyrzhanov, M. Rizig, A. Aitkulova, N. Zharkinbekova, C. Shashkin, G. Kaishibayeva, et al., Parkinson's disease in central Asian and Transcaucasian countries: a review of epidemiology, genetics, clinical characteristics, and access to care, Parkinsons Dis. 2019 (2019) 2905739.
[22] P. Lindgren, S. von Campenhausen, E. Spottke, U. Siebert, R. Dodel, Cost of Parkinson's disease in Europe, Eur. J. Neurol. 12 (Suppl. 1) (2005) 68–73.
[23] J.H. Siddiqui, D. Bhatti, F. Alsubaie, J.A. Bajwa, Movement disorders and deep brain stimulation in the Middle East, World Neurosurg. 113 (2018) e314–e319.
[24] K. Orayj, E. Lane, Patterns and determinants of prescribing for Parkinson's disease: a systematic literature review, Parkinsons Dis. 2019 (2019) 9237181.